# Optimized Trajectory Planning and Control in a 4-DOF Serial Manipulator

**PROJECT REPORT**

***Submitted by***


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***Under the guidance of***


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**SRM INSTITUTE OF SCIENCE AND TECHNOLOGY**

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# BONDAFIDE CERTIFICATE

Certified that this project report titled "**OPTIMIZED TRAJECTORY PLANNING AND CONTROL IN A 4-DOF SERIAL MANIPULATOR**" is the Bonafide work of "**BHARATHWAJ MANOHARAN [RA2512053010003], VANDHIYA DEWAN [RA2512053010014], MOHAMAD RIZVAN [RA2512053010015]**" who carried out the project work under my supervision as a batch. Certified further, that to the best of my knowledge the work reported herein does not form any other project report on the basis of which a degree or award was conferred on an earlier occasion for this or any other candidate.

Date: Project Supervisor:

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# ABSTRACT

This paper presents an optimization strategy for trajectory planning and control in a low-cost, 5-DOF serial manipulator, built with 3D-printed parts and driven by hobby-grade servos (MG995, SG90). The baseline system, using open-loop setpoint control via Wi-Fi, suffers from significant tracking errors, jitter, and instability due to hardware limitations, including non-real-time PWM, power supply issues, and actuator nonlinearities like backlash and deadband. A kinematic model based on the Denavit-Hartenberg (D-H) convention is developed, incorporating forward and inverse kinematics along with Jacobian analysis. The optimization replaces the baseline control with a hardware-aware pipeline, combining a quintic polynomial trajectory planner with a kinematic velocity-feedforward control law. This approach, computationally feasible on the ESP8266, reduces actuator current spikes and improves trajectory smoothness. Experimental results show significant reductions in RMS tracking error, peak error, and mechanical jerk compared to the baseline system. While optimization improves performance, the precision of the manipulator is ultimately limited by the nonlinearities of the low-cost actuators. This work demonstrates that software optimization can notably enhance performance in affordable robotic systems, despite inherent hardware constraints.

# TABLE OF CONTEXT

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